Homework

Homework 4 — A maximum and an equation — 60 min.

Consider the function f defined on $[0, +\infty)$ by $f(x) = (x+1)e^{-x}$. Its graph is denoted $\mathcal{C}_f$.

1. Calculate $f(0)$ and the limit of f at $+\infty$. Calculate $f'(x)$, then draw up the table of variations of f.
2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of the possible point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has a unique solution α on $[0, +\infty)$.
4. Write a Python function that uses bisection on $[0, 2]$ and returns two real numbers a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with abscissa 0. Is the position of the curve relative to this single tangent enough to state that f is concave on its whole domain? Justify.